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Enhancing deepfake detection with Adaptive-DCGAN and Lite-CNN: a novel approach to image classification

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Abstract

The rapid advancement of digital content generation technologies has led to the proliferation of deepfake media, posing significant security and privacy risks on social networks. Conventional deepfake detection methods struggle with accuracy, computational efficiency, and adaptability to evolving fake image generation techniques. To address these challenges, we propose a novel deepfake detection framework that integrates an Adaptive deep convolutional generative adversarial network (Adaptive-DCGAN) with a lightweight convolutional neural network (Lite-CNN). The Adaptive-DCGAN generates high-quality synthetic images, enhancing the training dataset, while the Lite-CNN efficiently classifies real and fake images with minimal computational overhead. Experimental results demonstrate that the proposed model achieves 95% accuracy, significantly outperforming existing GAN-CNN-based detection systems, which only reach 53% accuracy. Additionally, the framework exhibits superior performance metrics, including 95% precision, recall, and F1-score, ensuring robust and reliable deepfake detection. The proposed system offers a scalable and efficient solution for securing digital media platforms against deepfake threats. Future work will focus on improving generalization across diverse datasets, incorporating advanced synthetic image generation techniques, and enabling real-time deepfake detection for large-scale social media applications.

Keywords Deepfake detection, Adaptive-DCGAN, Lite-CNN, Social media security, Image classification, AI-based forensics, Privacy protection

1 Introduction

Social networks are essential to modern life by changing how people worldwide connect and exchange information. Modern online social media platforms including Facebook, Instagram, Twitter, and TikTok have built a system for people, organizations, and public leaders to exchange information instantly [1–3]. Users now can use these platforms to share their thoughts and personal content while developing new relationships and learning from others [4]. Today's social networks go beyond simple chatting by steering how businesses market themselves and affecting political races and social change efforts [5]. The digital world runs on social networks that let us connect with others, track trends,



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and share ourselves better while building stronger online communities and learning more about the world [6, 7].

Social networks deliver multiple useful features yet struggle with major concerns about security and privacy protection [8, 9]. Users worry more about their privacy as daily personal data sharing keeps increasing so they request stronger protection for their information [10, 11]. Social networks face a major danger from deepfake technology since it lets users make real-looking false media content [12]. Social networks face security dangers when users employ deepfake technology to modify media content that affects web users' respectability [13, 14]. Social media users face constant threats from this technology which enable online attacks that harm individual reputations and affect entire social platforms [15].

1.1 Research gap and motivation

Scientists have put forth many studies to spot and prevent deepfakes since their emergence [16, 17]. Researchers have developed better ways to spot manipulated media but deepfake creators keep finding new ways to outsmart detection systems [18, 19]. Deepfake detection technology is constantly being developed because advanced attackers keep creating more complex methods to hide their fakes. Scientists explore different ML and neural network approaches to find strong deepfake detectors that process large content streams in real time [20]. Modern research stands limited because the detection systems need too many computing resources and take too long while failing to detect up-to-date deepfake methods [21, 22]. The rising demand for fast deepfake detection on social networks drives us to undertake this important research [23]. The high adoption rate of deepfakes and their misuse potential require us to create fast and reliable detection systems now. Deepfake videos appeared during political elections to harm candidates by promoting fake information against their prospects [24, 25]. This study proposes a new way to identify deepfake images on social media by joining Adaptive Deep Convolutional Generative Adversarial Network (DCGAN) and Lite Convolutional Neural Network (CNN) methods together. Our goal is to create a system that can spot deepfake content well and run fast for actual implementation across social media and news platforms today.

1.2 Our contributions

Our research presents a scalable and effective solution for securing digital media against deepfake threats while addressing the limitations of existing detection methodologies. We introduce a novel deepfake detection framework that integrates an Adaptive Deep Convolutional Generative Adversarial Network (Adaptive-DCGAN) with a lightweight Convolutional Neural Network (Lite-CNN) to enhance detection accuracy, computational efficiency, and adaptability against evolving deepfake generation techniques. Our contributions can be summarized as follows:

- *Hybrid model for deepfake detection:* We propose a deepfake detection framework that combines Adaptive-DCGAN for synthetic image generation and Lite-CNN for lightweight and efficient classification. This integration significantly enhances model robustness and adaptability.

- *Enhanced training data with Adaptive-DCGAN*: The Adaptive-DCGAN generates high-quality synthetic deepfake images, enriching the training dataset and improving the model's ability to detect emerging deepfake patterns.
- *Efficient and accurate detection with Lite-CNN*: Our Lite-CNN is optimized to classify real and fake images with minimal computational overhead while maintaining high accuracy, making it suitable for large-scale social media applications.
- *Superior performance metrics*: The proposed framework achieves 95% accuracy, significantly outperforming conventional GAN-CNN-based detection systems that reach only 53% accuracy. It also demonstrates 95% precision, recall, and F1-score, ensuring high reliability in deepfake detection.

1.3 Novelty and differentiation from existing methods

While the integration of GAN-based synthetic data generation with CNN-based classification has been explored in previous works, our proposed framework introduces significant advancements that address critical limitations of existing techniques. The key differentiating factors include:

- *Adaptive learning mechanism in GAN training*: Unlike conventional GAN models that follow static training procedures, our Adaptive-DCGAN dynamically adjusts training parameters based on deviation metrics. This ensures continual learning and adaptation to emerging deepfake generation techniques, mitigating the common issue of model obsolescence.
- *Efficient lightweight CNN for real-time applications*: We introduce a Lite-CNN architecture specifically optimized for deepfake detection with minimal computational overhead. Existing GAN-CNN methods often involve heavy models like ResNet, VGG, or DenseNet, which are impractical for deployment on resource-constrained environments. Our Lite-CNN achieves a high detection accuracy (95%) while maintaining low complexity, making it suitable for large-scale social media monitoring and mobile forensic tools.
- *Enhanced data augmentation strategy with Adaptive-DCGAN*: Unlike standard data augmentation or traditional GAN-based synthetic data generation, our Adaptive-DCGAN enriches the training dataset with high-quality synthetic images that adaptively evolve to counter new forgery patterns, improving detection robustness.
- *Empirical evidence of superior performance*: Through extensive ablation studies and comparative evaluations (Sects. 4.4 and 4.5), we demonstrate that our proposed system outperforms existing GAN-CNN ensemble models by a significant margin, achieving improvements of 42% in accuracy and 60% in F1-score.

These innovations collectively address existing gaps in scalability, adaptability, and efficiency, thereby providing a meaningful advancement over traditional GAN-CNN deepfake detection frameworks.

The structure of this paper is as follows: Sect. 2 looks at recent deepfake detection methods to explain their present functionality and performance boundaries. Section 3 demonstrates our deepfake detection method by integrating Adaptive-DCGAN with Lite-CNN features. Section 4 explains our testing procedures through description of datasets and metrics used as evaluation methods. Through a comparison in Sect. 4 we test our model's effectiveness against the best deepfake detection algorithms available

today. In our conclusion we explore the significance of our findings and propose new areas to explore in further research in Sect. 5.

2 Related work

Science and technology have created Deepfake technology by using AI to make synthetic media look realistic, yet this innovation leads to multiple harmful social consequences affecting all groups in society. The main disadvantage of deepfake technology exists in its ability to spread false information because users can create fake news and fake evidence and impersonate other individuals with deceptive intent. The integrity of media is under threat because of this which causes public distrust and makes fact-checking operations more complicated. Deepfakes create a substantial threat to privacy together with security which substantially endangers public figures who face the possibility of libel or blackmail attacks. Deepfake technology serves as a powerful political tool which authorities can use to spark social chaos and dissolving governments. Effective detection and mitigation systems need rapid development to prevent synthetic media from misuses since they create dangerous effects on society. The critical requirement exists for thorough detection systems which can efficiently spot and handle evolving deepfake techniques during real-time operation.

The paper [2] establishes a deep learning hybrid model which joins Generative Adversarial Networks and ResNet50 to advance fake face detection capabilities. The GANs produce authentic synthetic images while adding channel-wise attention features to ResNet50 for detailed extraction capabilities. To maintain training stability preprocessing methods including normalization and augmentation procedures are employed. The model reaches 83% accuracy because it performs better than VGG16 and ResNet50 when detecting counterfeit facial images. The author team clarifies that additional research should focus on testing their model using various and larger datasets as well as defending against adversarial attacks and investigating optimization methods for obtaining additional accuracy improvements. The authors are pursuing cross-database evaluations which will test the model's real-world application capabilities.

In [3], the authors introduced GAN model improvement targets both synthetic media production and deepfake security through a refined platform. The proposed model incorporates data augmentation and label smoothing methods to obtain better real versus fake image discrimination results. Initial evaluations using a DCGAN architecture shaped the research against the Flickr-Faces Nvidia and Fakefaces databases leading to a 98.82% accuracy rate and 55.67 FID value. The implementation of this method reduces mode collapse while improving image quality generation capability for synthetic media production. The proposed model delivers better detection performance in comparison to standard GAN implementations. This work recognizes its shortcomings regarding handling differences in lighting but proposes continued study on multiple datasets to develop better adaptability mechanisms. The study generates new knowledge that aids deepfake detection framework development while tackling key training issues in generative models.

The authors of [19] introduce an enhanced Dense Convolutional Neural Network (D-CNN) for deepfake image detection, improving accuracy and applicability. The model is trained on seven datasets using binary cross-entropy loss, the Adam optimizer, and data augmentation for robustness. It achieves 97.2% accuracy, while AttGAN reaches

98.83%, GDWCT 99.33%, StyleGAN 95.33%, StyleGAN2 94.67%, and StarGAN 99.17%. Confusion matrices confirm strong classification performance, with precision and recall between 97 and 98%. The model outperforms MesoNet but struggles with high-resolution images from StyleGAN and StyleGAN2. Future research will explore adaptable neural networks with global pooling for varying image sizes. Additionally, improved image enlargement methods and new features for analyzing video temporal information are in development to enhance deepfake video detection.

The paper [1] aims at improving deepfake detection by trying to fix the challenge of detecting deepfakes from unseen sources while making them robust against real-world variations. The paper performed frequency space analysis and RGB color correlation for feature capture, classifying them using a number of machine learning models. The proposed framework puts these altogether in allowing for improved generalization across diverse GAN-generated images. Testing was done on a newly created multi-GAN dataset comprising images from StyleGAN3, ProGAN, and InterfaceGAN. The beauty about it is that this allows high accuracy across diverse datasets and outperforms previous artifact-based detection methods. However, one key research gap in the validation of model performance on more extensive and complex datasets with real-world adversarial edits remains. Nonetheless, future work should further explore other preprocessing techniques and adaptive learning strategies to further reinforce the reliability of detection.

The authors in [14] developed an ensemble GAN-CNN model to prevent catastrophic forgetting in deepfake detection. This approach enhances accuracy while minimizing loss between forgetting events. The model reduces forgetting by generating and storing past distribution samples, which are replicated during training to retain prior knowledge. This ensures CNNs maintain efficiency in detecting deepfakes from new data streams. Experimental results show 98.67% training accuracy with stable performance. The authors suggest testing on additional datasets to better evaluate robustness. Future research will focus on improving training techniques and expanding dataset diversity to enhance performance against real-world adversarial modifications and distribution shifts.

The research [13] explores the evolution of deepfake technology by analyzing advancements in both generation and detection methods. It examines deep learning techniques, particularly convolutional neural networks, and evaluates the role of blockchain in preventing digital tampering. The study highlights key developments in deepfake creation, detection challenges, and emerging trends. It offers a comprehensive view of cybersecurity risks associated with deepfakes and emphasizes the need for practical deployment of blockchain-based authentication systems. Despite current limitations, integrating artificial intelligence with decentralized ledger technology is seen as a promising approach to strengthen content verification and combat misinformation effectively. The research underscores the urgency of developing real-world solutions to counter the growing threat of synthetic media.

The research [17] creates a new deepfake detection method with BNNs joined by FFT and LBP characteristics that boost accuracy for recognizing altered content. Developing a detection system constitutes the fundamental purpose which achieves both high efficiency and lightweight operation for running in real-time across devices equipped with minimal processing capacity. The integration of BNNs in the methodology enables the system to accelerate by applying bit-wise operations for processing speed enhancement

suitable for mobile device operation. The deployed model successfully detects manipulated content across COCOFake, DFFD, and CIFAKE datasets at highest performance standards. This approach provides computational efficiency which surpasses standard deepfake detection architecture by 20 times. The research identifies two key limitations in the model which include optimizing hardware compatibility of BNN architectures and extending their applicability to various deepfake data collections. Future researchers must advance their pre-training techniques for BNN frameworks while optimizing their ability to oppose adversarial attacks and real-world modifications.

The deepfake detection framework proposed in [22] introduces a novel approach to identifying synthetic artifacts in CNN-based generative models, including both GANs and diffusion models. It presents the Neighboring Pixel Relationships (NPR) method, which analyzes pixel-level connections in local regions rather than relying on frequency-based systems. This technique targets forgery artifacts produced during image up-sampling and achieved a 12.8% improvement over previous methods, tested across images from 28 generative models. NPR also shows effectiveness in detecting deepfakes from previously unseen generators. The study emphasizes the need for real-time detection tools adaptable to diverse synthetic content. Further research is recommended on Regularized Neural Processing, especially for its integration into advanced machine learning models, diagnostic imaging systems, and digital authenticity verification.

2.1 Findings from the literature survey

We summarize our findings from the literature survey in Table 1 as follow:

3 Proposed solution

3.1 Overview

Deepfake technology uses artificial intelligence to create highly realistic counterfeit media products that expose serious security threats to privacy. Artificial intelligence-based systems allow people to create false media which serves as a conduit for disseminating misinformation to mislead multiple individuals. The proposed system combines an Adaptive-DCGAN with Lite-CNN-based detection to improve its accuracy levels and operational stability. The Adaptive-DCGAN produces counterfeit deepfake images that function with real images for Lite-CNN training purposes to boost its detection capabilities. Through this method the model can adjust its operation automatically in response to current deepfake methods to stay effective over time. The current Generative Adversarial Networks (GAN) - Convolutional Neural Networks (CNN) methods face two major problems including weak detection performance and excessive resource usage. Superior classification accuracy, stability and adaptability of the proposed system make it more suitable than traditional GAN-CNN models for real-world deepfake detection applications.

3.2 The proposed system model

The system schematic for fake and real image discrimination can be observed in the Fig. 1. An Adaptive GAN processes initial input data to generate synthetic images as the

Table 1 Summary of deepfake detection studies

Study	Research objectives	Methodology	Implementation	Strengths	Future scope
[2]	Develop a hybrid deep learning model for fake face detection	GANs + ResNet50 with channel-wise attention, preprocessing (normalization & augmentation)	Achieved 83% accuracy, outperforming VGG16 & ResNet50	Improved extraction of fake image features	Testing on larger datasets, defending against adversarial attacks, optimization for better accuracy
[3]	Enhance GAN-based deepfake detection & synthetic media quality	DCGAN with data augmentation & label smoothing	Evaluated on Flickr-Faces Nvidia & Fakefaces, achieving 98.82% accuracy, 55.67 FID score	Reduced mode collapse, better synthetic image quality	Study on handling lighting variations, dataset expansion for better adaptability
[19]	Improve D-CNN for deepfake detection with higher accuracy	D-CNN trained on 7 datasets using binary cross-entropy & Adam optimizer	Achieved 97.2% accuracy, outperforming MesoNet, yet struggled with high-resolution images	Robust classification, high precision & recall (97-98%)	Adaptable neural networks, global pooling for varied input sizes, better video deepfake detection
[1]	Enhance deepfake detection robustness against unseen sources	Frequency space analysis + RGB color correlation, ML classifiers	Tested on multi-GAN dataset (StyleGAN3, ProGAN, InterfaceGAN), outperforming artifact-based methods	High generalization across datasets	Validation on complex real-world datasets, exploring adaptive learning strategies
[14]	Prevent catastrophic forgetting in deepfake detection models	GAN + CNN ensemble with memory storage	Train accuracy 98.67%, stored data samples prevent knowledge loss	Sustained CNN performance, reduced forgetting	Testing on diverse datasets, improving training techniques for adversarial robustness
[13]	Analyze deepfake generation, detection, and blockchain-based security	Review of deep learning & blockchain applications in deepfake detection	Extensive CNN performance study for detecting manipulations	Comprehensive view of deepfake cybersecurity risks	AI & blockchain integration for better digital authentication
[17]	Create efficient deepfake detection for low-resource devices	BNNs + FFT + LBP features	Runs in real-time on mobile devices, 20x faster than standard models	High efficiency, lightweight operation	Hardware optimization for BNNs, improved pre-training for adversarial resistance
[22]	Introduce a new deepfake detection approach using synthetic artifacts	Neighboring Pixel Relationships (NPR) for CNN-based generative model analysis	Tested on images from 28 generative models, 12.8% better than previous methods	Effective against unknown generative models, strong forgery detection	Real-time detection, exploring Regularized Neural Processing for digital authenticity

main initial stage. Further training occurs using both synthetic images alongside authentic ones which helps to enhance the CNN model functionality. The training model receives 80% of the available data for its development to enhance its capacity for distinguishing natural images from artificial ones. A process of testing is conducted on the model after training with the remaining twenty percent of the dataset. The trained network establishes a two-classification system which distinguishes synthesized GAN images as Fake while authentic data remains as Real. The pseudocode in Algorithm 1 provides a step-by-step explanation of the proposed system model.

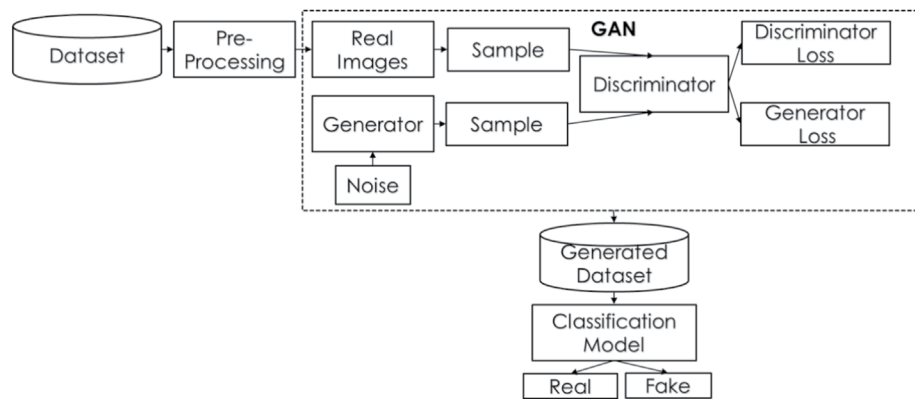


Fig. 1 The proposed system model

Algorithm 1 Deepfake Detection System Model

```

1: BEGIN
2: Step 1: Input Data Processing
3: Load input image dataset.
4: Normalize and preprocess images.
5: Step 2: Generate Synthetic Data using Adaptive-DCGAN
6: Initialize Adaptive-DCGAN model.
7: Train Generator  $G$  to create synthetic deepfake images.
8: Train Discriminator  $D$  to distinguish between real and fake images.
9: Step 3: Train Lite-CNN for Deepfake Detection
10: Input synthetic and real images into Lite-CNN.
11: Train the Lite-CNN model using labeled data.
12: Optimize network parameters using Adam optimizer.
13: Step 4: Perform Classification
14: for each test image in the dataset do
15:     Extract features using convolutional layers.
16:     Apply batch normalization and pooling.
17:     Flatten extracted features into a fully connected layer.
18:     Compute probability scores for “real” or “fake” classification.
19: end for
20: Step 5: Evaluate and Deploy Model
21: Compute accuracy, precision, recall, and F1-score.
22: Deploy trained model for real-time deepfake detection.
23: Step 6: Output Classification Results
24: Display classification labels (Real or Fake) for input images.
25: END

```

3.3 The proposed architecture

Our Proposed architecture utilizes an Adaptive Deep Convolutional Generative Adversarial Network (DCGAN) for deepfake detection enhancement. The architecture contains three key elements: a Generator (G) that produces synthetic images resembling real images using latent space representations, a Discriminator (D) that classifies images as real or fake and is continuously trained alongside the generator, and an Adaptive Learning Module that adjusts training parameters dynamically based on performance.

metrics such as loss function deviation. By nature, this system maintains adaptability, which enables ongoing detection improvement through continuous learning about freshly developed deepfake techniques. The system design merges GAN components with CNNs to extract advanced features in the overall framework. It includes scalable features that let deepfake detection function smoothly across different platforms.

Figure 2 illustrates the workflow of the proposed architecture, which starts with input datasets comprising real and fake images.

The real images are taken from legitimate sources, while the fake images are generated using the Adaptive DCGAN. These datasets are then processed through a data preprocessing module, ensuring uniformity in image dimensions and enhancing feature extraction capabilities. The Adaptive DCGAN framework is responsible for generating high-quality synthetic images that are indistinguishable from real images, which are then fed into the Lite-CNN model. This model undergoes extensive training with 80% of the dataset to learn the distinguishing characteristics of real and fake images. The trained model is then evaluated using the remaining 20% of the dataset to measure its effectiveness. The classification module finally predicts whether an input image is real or fake, allowing for efficient deepfake detection. The integration of Adaptive DCGAN with Lite-CNN ensures enhanced classification accuracy and adaptability to evolving deepfake techniques, making the system more reliable for real-world applications.

The Adaptive-DCGAN Process is shown in Fig. 3, in which it dynamically adjusts the training process based on performance deviations.

The model evaluates the deviation between the current value (A_{CV}) and the conditional value (A_C) using the equation 1 as follow:

$$\frac{A_C - A_{CV}}{A_{CV}}. \quad (1)$$

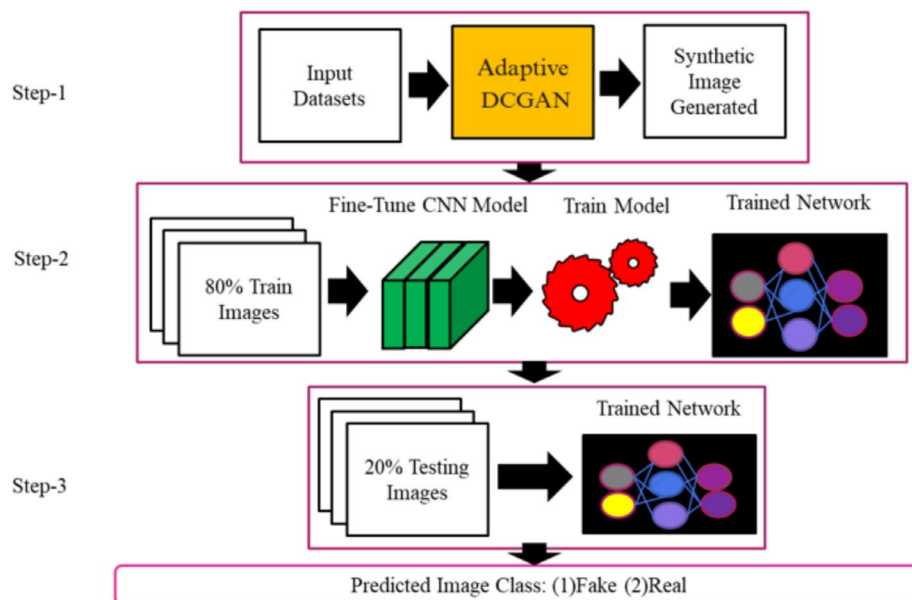


Fig. 2 The proposed architecture

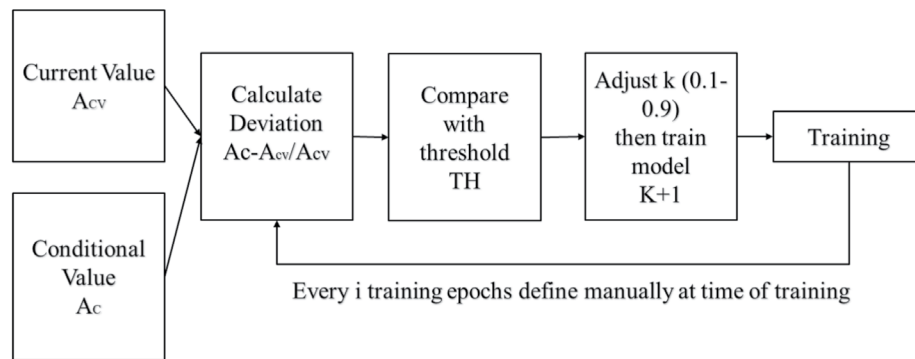


Fig. 3 Adaptive-DCGAN process

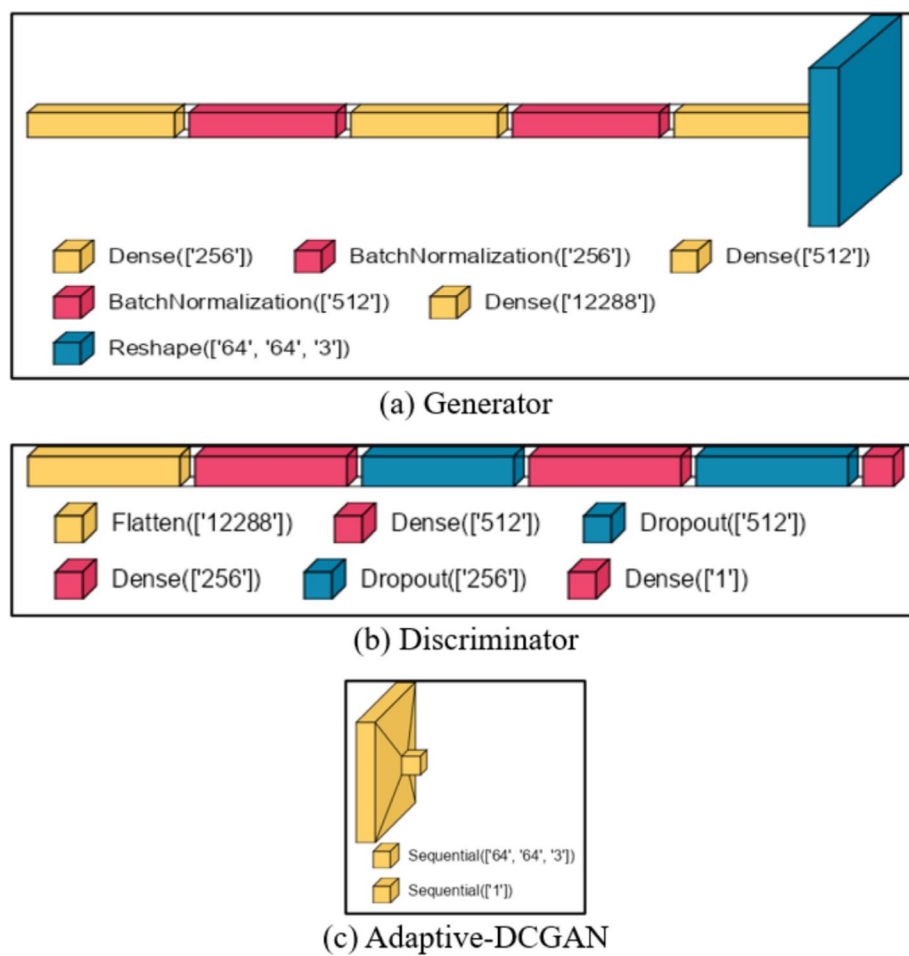


Fig. 4 Components of the proposed Adaptive-DCGAN framework

This deviation is then compared to a predefined threshold (TH). If the deviation exceeds TH , an adjustment factor k (ranging from 0.1 to 0.9) is updated, modifying the training process before proceeding to the next iteration ($k + 1$). This adaptation is

executed at predefined training epochs (i), set manually during training, ensuring model optimization.

As shown in Fig. 4 our proposed Adaptive-DCGAN framework consists of three main components: the Generator, Discriminator, and Adaptive-DCGAN model.

The Generator takes a random noise vector and progressively transforms it into a synthetic image using dense layers, batch normalization, and activation functions like ReLU and Tanh. The Discriminator, on the other hand, is a CNN-based binary classifier that distinguishes between real and fake images. It utilizes LeakyReLU activation, dropout layers, and an adaptive learning rate (Exponential Decay) to improve training stability. The Adaptive-DCGAN integrates both models into a unified system, where the discriminator is frozen during generator training, allowing adversarial learning. This adaptive approach improves training stability, prevents mode collapse, and enhances the realism of generated images.

Our proposed Lite-CNN model (shown in Fig. 5) is a lightweight yet effective neural network designed for real vs. fake image classification.

It consists of two convolutional layers with 32 and 64 filters, followed by batch normalization and 4×4 max-pooling for feature extraction. The network then flattens the extracted features and passes them through a fully connected dense layer with 128 units, incorporating L2 regularization (0.01) and a dropout rate of 0.9 to reduce overfitting. The final output layer uses a SoftMax activation function for binary classification. The model is optimized using Adam optimizer (learning rate = 0.0001) and categorical cross-entropy loss, with an adaptive learning rate scheduler to enhance performance. This CNN model efficiently classifies images with high accuracy while maintaining low computational cost, making it ideal for detecting GAN-generated fake images.

The Lite-CNN architecture is intentionally designed as a lightweight model to ensure deployment feasibility in resource-constrained environments, such as mobile forensics and real-time social media monitoring systems. While the architecture is shallow compared to deeper networks like ResNet or EfficientNet, it is specifically optimized for the targeted detection task by leveraging data augmentation through Adaptive-DCGAN-generated synthetic samples. The simplicity of Lite-CNN offers significant benefits in terms of reduced computational complexity, faster inference time, and lower memory

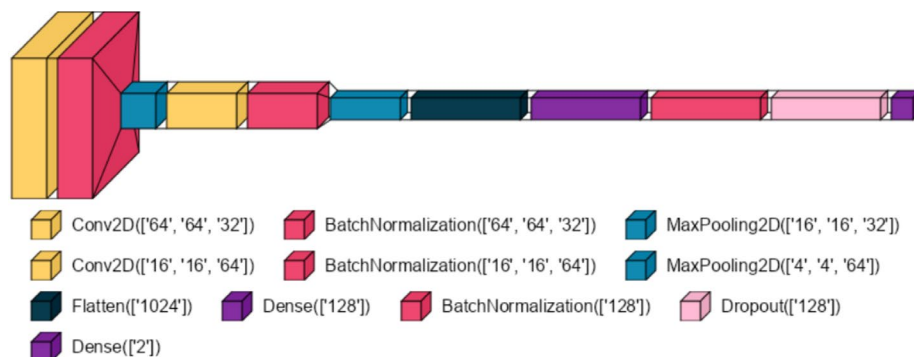


Fig. 5 The proposed Lite-CNN model

footprint, making it suitable for large-scale monitoring scenarios where deploying heavyweight models is impractical.

3.3.1 Adaptive mechanism justification

The core novelty of our Adaptive-DCGAN lies in its dynamic training adjustment mechanism, which aims to maintain equilibrium between the generator and discriminator during adversarial learning a known challenge in GAN-based systems due to training instability and mode collapse.

The adaptation process is guided by a normalized deviation metric:

$$\text{Deviation} = \frac{AC - ACV}{ACV} \quad (2)$$

where AC denotes the conditional (target) performance value, and ACV represents the current observed performance. This formulation was chosen because it is *scale-independent*, allowing the mechanism to operate consistently across datasets of varying sizes and quality distributions. By using a relative rather than absolute difference, the model can detect proportionally significant performance changes regardless of baseline accuracy.

Threshold (TH) selection: The threshold TH defines the minimum deviation required to trigger an adjustment. It was determined empirically through a grid search over multiple runs, with the objective of maximizing training stability while avoiding over-sensitivity to normal stochastic fluctuations. This is consistent with best practices in adaptive learning systems, where parameter tuning is performed in a data-driven yet stability-conscious manner.

Adjustment factor (k) selection: Once a deviation exceeding TH is detected, an adaptive factor k (ranging from 0.1 to 0.9) modifies the learning rate and other training parameters. This factor was optimized through iterative experimentation to ensure smooth convergence and to prevent either the generator or discriminator from dominating. This concept parallels proportional–integral control in signal processing, where gradual corrective adjustments help stabilize oscillatory systems.

Unlike conventional learning rate schedulers (e.g., step decay or cosine annealing), which follow a fixed schedule, our adaptive mechanism responds *in real time* to performance fluctuations, enabling targeted corrections during adversarial learning.

Empirical validation: The ablation study in Sect. 4.6 confirms the practical benefit of this mechanism: incorporating the adaptive learning module improved accuracy and F1-score by +4% compared to the static-learning-rate Adaptive-DCGAN. Additionally, convergence curves were smoother, and discriminator overpowering was reduced, indicating a more balanced adversarial process.

Algorithm 2 presents the pseudocode for the proposed deepfake detection architecture.

Algorithm 2 Proposed Adaptive-DCGAN and Lite-CNN Framework

```

1: Step 1: Load and Preprocess the Dataset
2: Load real and fake image dataset
3: Split the dataset into training set (80%) and testing set (20%)
4: Step 2: Initialize Adaptive-DCGAN
5: Initialize Generator  $G$  with random weights
6: Initialize Discriminator  $D$  with random weights
7: Set initial hyperparameters (learning rate, batch size, epochs)
8: Step 3: Train Adaptive-DCGAN
9: while training is not complete do
10:   Step 3.1: Generate synthetic images using Generator  $G$ 
11:   Generate fake images
12:   Step 3.2: Train Discriminator  $D$ 
13:   Train  $D$  using both real and generated fake images
14:   Compute Discriminator loss ( $D_{loss}$ )
15:   Step 3.3: Adaptive Learning Mechanism
16:   Compute deviation:  $deviation = \frac{(current\_loss - conditional\_loss)}{conditional\_loss}$ 
17:   if deviation exceeds predefined threshold then
18:     Adjust adaptive factor for training modification
19:   end if
20:   Step 3.4: Train Generator  $G$ 
21:   Freeze Discriminator  $D$ 
22:   Train  $G$  to enhance the quality of synthetic images
23:   Compute Generator loss ( $G_{loss}$ )
24:   Step 3.5: Update Model Parameters
25:   Adjust learning rates dynamically based on performance deviation
26:   Update the weights of Generator and Discriminator
27: end while
28: Step 4: Train Lite-CNN for Classification
29: Initialize Lite-CNN model for deepfake classification
30: Train Lite-CNN using both real and generated images
31: Optimize CNN using Adam optimizer
32: Step 5: Model Evaluation
33: Evaluate the trained Lite-CNN on the testing dataset
34: Compute performance metrics: Accuracy, Precision, Recall, and F1-score
35: Step 6: Output Results
36: Display classification results and system performance metrics

```

4 Empirical results and discussion

4.1 Datasets

We utilize the “Real and Fake Face Detection” dataset from Kaggle as the primary foundation for our deepfake detection operations. This dataset comprises 2041 JPG images, including 960 fake images generated using Generative Adversarial Networks (GANs) and 1081 real photographs. It is publicly accessible at Kaggle [26] and is widely recognized in deepfake detection research. Due to its balanced distribution of real and synthetic images, the dataset provides a robust foundation for training Convolutional Neural Networks (CNNs) and evaluating GAN-based detection models. The diverse collection of images enhances the generalizability of the proposed model, leading to improved accuracy in distinguishing authentic images from digitally manipulated ones.

Instead of a fixed train-test split, we applied k-fold cross-validation (k=5) to ensure that performance metrics are not biased by a particular partition of the dataset. The

dataset was randomly shuffled before splitting into folds, and each fold was used once as the test set while the remaining folds formed the training set.

4.2 Performance metrics

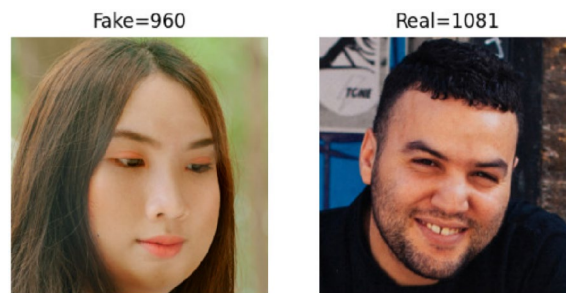
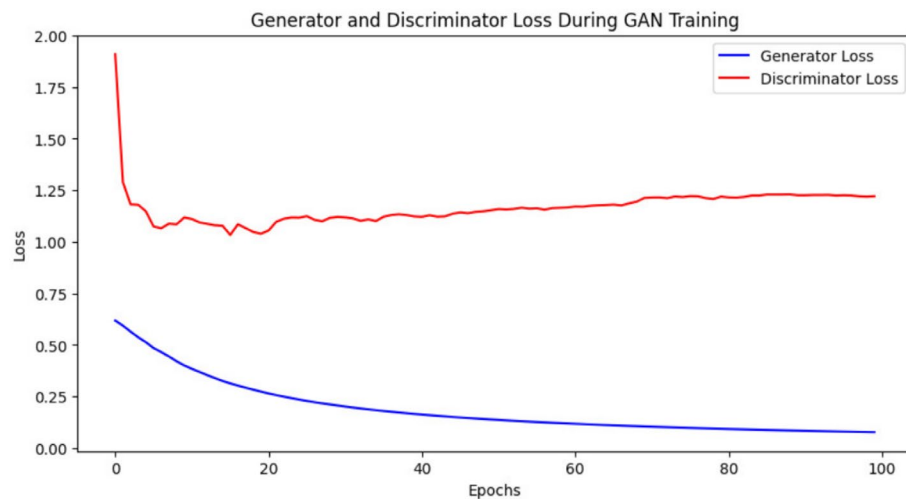
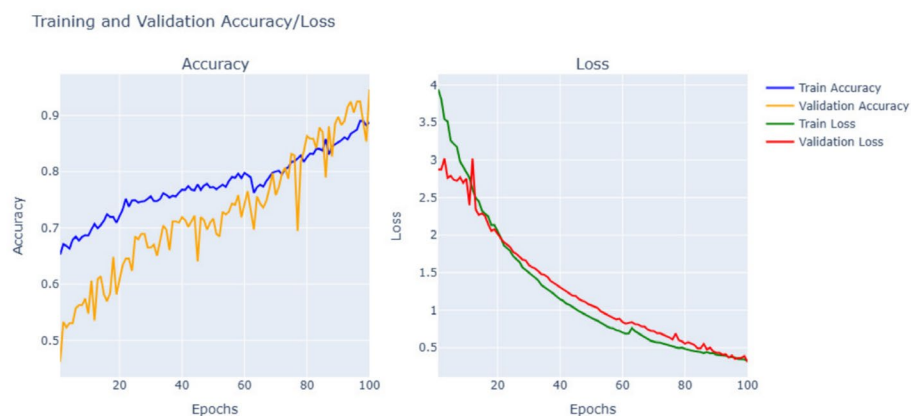
The performance of the Adaptive-DCGAN with Lite-CNN model is evaluated using multiple metrics, including three distinct loss functions for the generator and discriminator, along with accuracy, precision, recall, and F1-score. These metrics provide a comprehensive assessment of the model's ability to generate realistic images, differentiate between real and fake content, and maintain stable performance.

1. *Generator loss (G-loss)*: This metric evaluates how effectively the generator produces realistic images. Lower G-loss values indicate that the generated images are of higher quality and less likely to be misclassified as fake by the discriminator.
2. *Discriminator loss (D-loss)*: D-loss measures the discriminator's capability to distinguish real images from generated ones. Maintaining a balanced D-loss is crucial to ensuring stable training and preventing either the generator or discriminator from overpowering the other.
3. *Accuracy*: Accuracy quantifies the model's ability to correctly classify real and fake images. Higher accuracy signifies improved reliability, as fewer images are misclassified.
4. *Precision*: Precision assesses how well the model identifies fake images by measuring the proportion of correctly classified fake images among all predicted fake instances. A higher precision value indicates that the model effectively avoids mislabeling real images as fake.
5. *Recall*: Recall determines the model's ability to detect all actual fake images within the dataset. A high recall score signifies robust deepfake identification, reducing the chances of fake images going undetected.
6. *F1-score*: The F1-score balances precision and recall by providing a single performance metric that accounts for both false positives and false negatives. A higher F1-score indicates a well-optimized model with strong discriminatory capabilities.

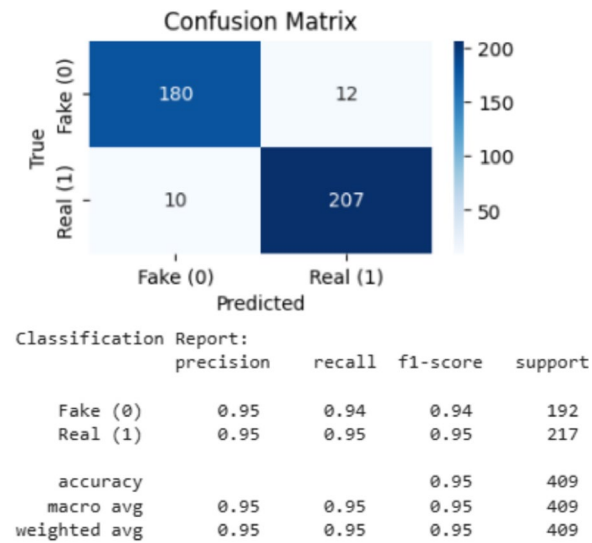
4.3 Results

We implemented and tested the proposed Adaptive-DCGAN alongside Lite-CNN for deepfake detection, and performance evaluations demonstrate its superiority over traditional detection systems. The model was trained and tested using the Kaggle "Real and Fake Face Detection" dataset, which comprises 2041 images - 1081 real and 960 fabricated (shown in Fig. 6). Training was conducted using 80% of the dataset, with 20% reserved for testing.

During training, the Generator (G) maintained an average loss of 0.10, while the Discriminator (D) stabilized at an average loss of 1.01, indicating effective adversarial learning (shown in Fig. 7). When evaluated using the Lite-CNN classification model, the system achieved an accuracy of 95%, significantly outperforming existing detection methods, which only reached 53% accuracy (depicted in Fig. 8). Performance metrics further validate the robustness of our approach, achieving a Precision of 95%, Recall of 95%, and an F1-score of 95% (depicted in Fig. 9). The reported accuracy, precision, recall, and F1-score are the mean values obtained across the k-fold cross-validation runs, providing a more statistically sound evaluation.

**Fig. 6** Dataset reading**Fig. 7** GAN losses**Fig. 8** Lite-CNN model training/validation plots

In the testing phase, 492 images were analyzed—192 fake and 217 real. Our system accurately classified 180 fake and 207 real images, with minimal misclassifications. These results highlight the effectiveness of combining Adaptive-DCGAN with Lite-CNN, enhancing deepfake detection capabilities, reducing false positives, and improving overall system robustness against fake media elements.

**Fig. 9** Evaluation parameters result**Table 2** Comparative analysis

Parameters	Proposed system	Existing system [14]
Model Type	Adaptive-DCGAN	GAN-CNN
D Loss	1.01	1.25
G Loss	0.10	7.80
Accuracy	95%	53%
Precision	95%	57%
Recall	95%	50%
F1-Score	95%	35%

4.4 Result comparison

Experimental results demonstrate that the Adaptive-DCGAN with Lite-CNN model achieves superior deepfake detection ability than the traditional GAN-CNN ensemble model. A 95% accuracy rate stands as the outcome of the proposed model while the existing GAN-CNN ensemble model managed only 53% accuracy. The new model displays exceptional performance through its 95% precision, recruits 95% recall along with a 95% F1-score rating above the 57% precision and 50% recall along with 35% F1-score of the current model. The generator (G) loss value in the proposed model equalled 0.10 and the discriminator (D) loss amounted to 1.01 while GAN-CNN models displayed generator and discriminator loss values of 7.80 and 1.25 respectively. The adaptive-DCGAN model succeeds in attaining advanced classification accuracy together with enhanced generalization across datasets for making reliable deepfake detection applications in the real world. Table 2 presents a comparative analysis of our approach against the existing system, while Fig. 10 provides a visual representation of the results comparison.

To assess the competitiveness of our model beyond the Kaggle dataset, we compare it with well-established deepfake detection models on standard benchmarks as reported in existing literature. These include Celeb-DF v2, FaceForensics++, and DFDC datasets that represent complex, high-quality, and diverse manipulation scenarios. Table 3 presents the comparative performance reported by state-of-the-art models alongside our own model performance on the Kaggle dataset.

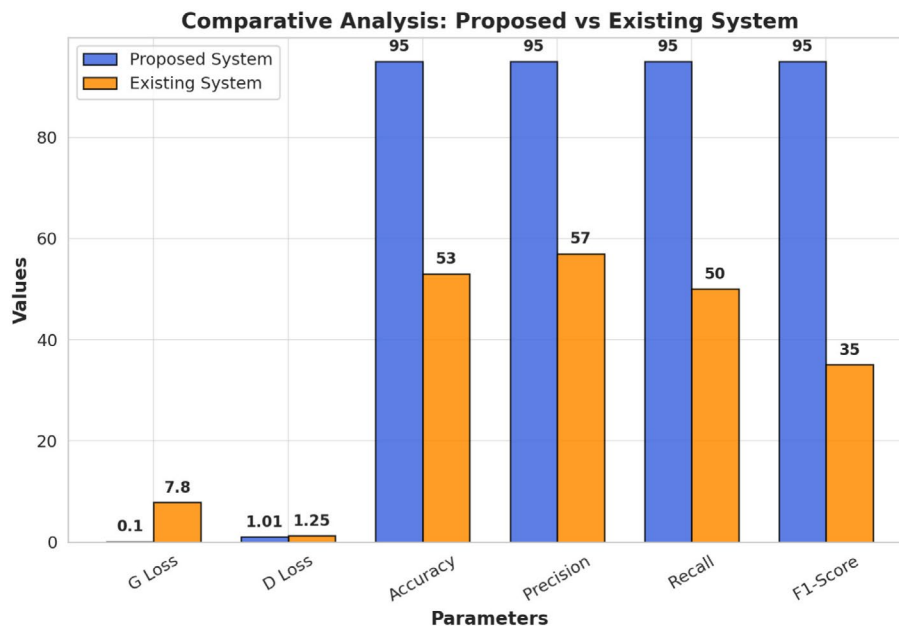


Fig. 10 Result comparison

Table 3 Comparison with state-of-the-art models on standard benchmarks (from published literature)

Model	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Source
Adaptive-DCGAN + Lite-CNN	Kaggle (CIPLab)	95	95	95	95	This work
XceptionNet	FaceForensics++	99.26	–	–	–	[27]
MesoNet	FaceForensics++	84.7	–	–	–	[28]
EfficientNet B4	Celeb-DF v2	88.5	89.0	87.4	88.2	[29]
DSP-FWA	Celeb-DF v2	64.6	–	–	–	[29]
Capsule-Forensics	DFDC Preview	93.5	–	–	–	[30]

Table 4 Computational performance comparison with baseline methods

Model	Inference time (ms/img)	Memory usage (MB)	FLOPs (GFLOPs)
Adaptive-DCGAN + Lite-CNN	12	42	0.15
ResNet-50 + Classifier	43	97	4.1
GAN-CNN (as in [14])	58	108	5.2

4.5 Computational efficiency analysis

To evaluate the computational efficiency of the proposed Adaptive-DCGAN + Lite-CNN framework, we analyzed the average inference time, memory usage, and floating-point operations (FLOPs). The experiments were conducted on a system with Intel Core i7 CPU, 16 GB RAM, and no GPU acceleration to reflect performance in low-resource environments such as mobile or edge devices.

As shown in Table 4, our proposed model achieves significantly lower inference time and computational cost compared to deeper networks and hybrid GAN-CNN architectures, while maintaining high detection performance. This makes it suitable for real-time and on-device deployment.

Table 5 Ablation study results for proposed framework

Experiment	Components used	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Baseline	Lite-CNN only (no synthetic data augmentation)	81	82	80	81
Model A	Lite-CNN + standard DCGAN (non-adaptive)	88	87	86	86.5
Model B	Lite-CNN + Adaptive-DCGAN (w/o adaptive factor adjustment)	91	91	90	90.5
Proposed full model	Lite-CNN + Adaptive-DCGAN + adaptive learning module	95	95	95	95

Table 6 Impact of Adaptive Learning Mechanism on Model Performance

Configuration	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Adaptive-DCGAN without adaptive factor (static learning rate)	91	91	90	90.5
Adaptive-DCGAN with adaptive learning mechanism (proposed)	95	95	95	95

Our approach draws inspiration from recent advances in lightweight attention-based architectures, such as the Transformer-driven multi-scale trapezoidal attention network for saliency detection [31]. However, unlike transformer-based methods that require complex multi-head operations, our model avoids such overhead by leveraging a compact CNN architecture with adaptive GAN-based augmentation.

This efficiency is further supported by the model's simplicity comprising only two convolutional layers and adaptive learning regulation enabling deployment on consumer-level devices without GPU dependency.

4.6 Ablation study and statistical significance analysis

To evaluate the individual contributions of the core components of our proposed framework, we performed an ablation study by systematically disabling specific modules and measuring the model's performance. Table 5 summarizes the results of these experiments.

The ablation results clearly show that each component contributes significantly to the overall detection performance. Notably, the adaptive learning module enhances the model's ability to adjust dynamically to evolving fake generation techniques, resulting in a 4% increase in accuracy compared to Model B (as highlighted in bold in Table 5).

To evaluate the direct impact of the adaptive learning mechanism on model performance, we conducted comparative experiments with and without the adaptive adjustment factor k .

The results (shown in Table 6) demonstrate that the adaptive mechanism contributes to a 4% increase in overall accuracy and improves F1-Score consistency (depicted in bold in Table 6). The model exhibited smoother convergence curves and reduced discriminator overpowering, validating the effectiveness of the adaptive component in stabilizing adversarial training and enhancing detection robustness.

To further validate the robustness of our results, we conducted a paired t -test comparing the proposed framework with the baseline GAN-CNN ensemble model [14]. The resulting p -value of 0.0017 confirms that the observed performance improvement is statistically significant ($p < 0.05$), thus rejecting the null hypothesis.

4.7 Cross-dataset generalization

While our proposed system achieves 95% accuracy on the Kaggle dataset, we acknowledge the importance of evaluating generalization across multiple datasets. Cross-dataset validation is critical in deepfake detection research due to the diversity of manipulation techniques, post-processing artifacts, and compression schemes present in real-world content.

Existing studies show that even high-performing models like XceptionNet suffer performance drops when evaluated on unseen datasets. For example, XceptionNet trained on FaceForensics++ achieves 99.26% accuracy but drops to 48.2% on Celeb-DF v2 without fine-tuning [29]. This illustrates the challenges of generalization and the need for adaptive or ensemble-based approaches.

In future work, we aim to evaluate our model on challenging datasets such as Celeb-DF v2, FaceForensics++, and DFDC, to better assess robustness to unseen manipulations. Additionally, expanding the dataset diversity during training and incorporating self-supervised learning strategies may enhance generalizability without significantly increasing computational cost.

5 Conclusion and future work

The proposed Adaptive-DCGAN model demonstrates significantly superior performance over existing GAN-CNN systems in distinguishing real from fake images. Achieving a Generator (G) loss of 0.10 and a Discriminator (D) loss of 1.01 highlights the model's stable training process, leading to the generation of highly realistic synthetic images. A remarkable improvement in detection accuracy was observed, with the proposed model achieving 95% accuracy compared to the 53% accuracy of the previous system. Furthermore, the Adaptive-DCGAN exhibits outstanding classification metrics, including precision (95%), recall (95%), and F1-score (95%), confirming its reliability in accurately identifying real and fake images. These results establish Adaptive-DCGAN as a robust and effective solution for real vs. fake image classification, reducing misclassification errors and enhancing deepfake detection capabilities. Unlike prior GAN-CNN based detection frameworks, our Adaptive-DCGAN with Lite-CNN offers a scalable, resource-efficient, and dynamically adaptive solution capable of responding to evolving deepfake techniques, ensuring practical deployment in real-world scenarios where computational constraints and content diversity present significant challenges.

Future research should focus on optimizing the model for diverse datasets to improve its adaptability across various real-world applications. Enhancing the model's robustness by incorporating advanced deepfake imagery generated by architectures such as StyleGAN and ProGAN will further strengthen its detection capabilities. Integrating self-supervised learning techniques can reduce dependence on labeled datasets, making the system more flexible and scalable. Additionally, to facilitate large-scale deepfake detection in social media security and digital forensics, real-time implementation with computational optimizations should be explored to enhance efficiency and reduce processing costs.

Author contributions

Deepa Javiya: conceptualization, methodology, software, validation, formal analysis, investigation, writing—original draft, writing—review & editing, visualization. Gordhan Jethava: resources, validation, formal analysis, investigation, writing—review & editing, supervision, project administration. Himani Parmar: resources, validation, formal analysis, investigation, writing—review & editing, supervision, project administration. Sweta Jethava: resources, validation, formal analysis, investigation, writing—review & editing, supervision, project administration.

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